



Student Number: \_\_\_\_\_

Teacher: \_\_\_\_\_

Class: \_\_\_\_\_

FORT STREET HIGH SCHOOL

## 2018

### HIGHER SCHOOL CERTIFICATE COURSE

### ASSESSMENT TASK 3: TRIAL HSC

# Mathematics

**Time allowed: 3 hours**

(plus 5 minutes reading time)

| Syllabus Outcomes | Assessment Area Description and Marking Guidelines                                                                                                                                                    | Questions  |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
|                   | Chooses and applies appropriate mathematical techniques in order to solve problems effectively                                                                                                        | 1-10       |
| H3                | Manipulates algebraic expressions involving logarithmic and exponential functions                                                                                                                     | 11         |
| H2                | Constructs arguments to prove and justify results                                                                                                                                                     | 13         |
| H5, H6, H8        | Applies appropriate techniques from the study of calculus, geometry, probability, trigonometry and series to solve problems. Uses the derivative to determine the features of the graph of a function | 12, 14, 15 |
| H4, H9            | Expresses practical problems in mathematical terms based on simple given models. Communicates using mathematical language, notation, diagrams and graphs                                              | 16         |

### Total Marks 100

#### Section I 10 marks

Multiple Choice, attempt all questions,

Allow about 15 minutes for this section

#### Section II 90 Marks

Attempt Questions 11-16,

Allow about 2 hours 45 minutes for this section

### General Instructions:

- Questions 11-16 are to be started in a new booklet
- The marks allocated for each question are indicated
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations.
- Marks may be deducted for careless or badly arranged work.
- Board – approved calculators may be used

| Section I  | Total 10 | Marks |
|------------|----------|-------|
| Q1-Q10     |          |       |
| Section II | Total 90 | Marks |
| Q11        | /15      |       |
| Q12        | /15      |       |
| Q13        | /15      |       |
| Q14        | /15      |       |
| Q15        | /15      |       |
| Q16        | /15      |       |
|            | Percent  |       |

## Section I

10 marks

Attempt Questions 1 to 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 to 10

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1. What is the locus of a set of points that is equidistant from a fixed point and a fixed line?
- (A) a parabola      (B) a hyperbola      (C) a circle      (D) a straight line
2. Which one of the following quadratic equations has two distinct real roots?
- (A)  $y = x^2 - 4x + 4$       (B)  $y = x^2 + 4x + 4$
- (C)  $y = x^2 - 4x - 4$       (D)  $y = x^2 + 4$
3. The solutions of  $\sqrt{3} \tan x = -1$  for  $0 \leq x \leq 2\pi$  are?
- (A)  $\frac{2\pi}{3}$  and  $\frac{4\pi}{3}$       (B)  $\frac{2\pi}{3}$  and  $\frac{5\pi}{3}$       (C)  $\frac{5\pi}{6}$  and  $\frac{7\pi}{6}$       (D)  $\frac{5\pi}{6}$  and  $\frac{11\pi}{6}$
4. Find the limiting sum of the geometric series  $\frac{2}{3} - \frac{2}{15} + \frac{2}{75} - \frac{2}{375} + \dots$
- (A)  $\frac{3}{5}$       (B) 0      (C)  $\frac{12}{15}$       (D)  $\frac{5}{9}$
5. Which of the following conditions for  $\frac{dP}{dt}$  and  $\frac{d^2P}{dt^2}$  describes the slowing growth of a variable B?
- (A)  $\frac{dP}{dt} > 0$  and  $\frac{d^2P}{dt^2} > 0$       (B)  $\frac{dP}{dt} < 0$  and  $\frac{d^2P}{dt^2} < 0$
- (C)  $\frac{dP}{dt} > 0$  and  $\frac{d^2P}{dt^2} < 0$       (D)  $\frac{dP}{dt} < 0$  and  $\frac{d^2P}{dt^2} > 0$

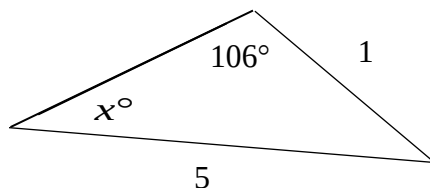
6. The quadratic equation  $x^2 + 4x - 1 = 0$  has roots  $\alpha$  and  $\beta$ .  
What is the value of  $\alpha\beta + (\alpha + \beta)$ ?

(A) 5                      (B) 3                      (C) -5                      (D) -3

7. If  $\ln a = \ln b + \ln c$ , then which of these is true?

(A)  $a = bc$               (B)  $a = b + c$               (C)  $\ln a = bc$               (D)  $a = \frac{b}{c}$

8. Which calculation gives the value of  $x$  in the diagram below?

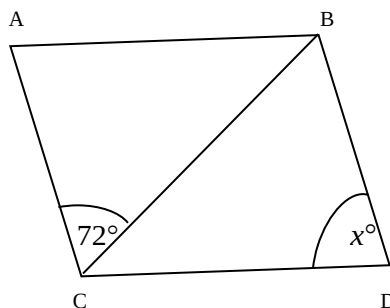


(A)  $x = \sin^{-1}\left(\frac{\sin 106^\circ}{5}\right)$               (B)  $x = \cos^{-1}\left(\frac{\sin 106^\circ}{5}\right)$   
(C)  $x = \sin^{-1}(5 \sin 106^\circ)$               (D)  $x = \sin^{-1}\left(\frac{1}{5}\right)$

9. A bag contains red and green lollies in the ratio of 7 : 2. If a lolly is selected at random, what is the probability that it is a green lolly?

(A)  $\frac{1}{7}$                       (B)  $\frac{7}{9}$                       (C)  $\frac{2}{7}$                       (D)  $\frac{2}{9}$

10. The quadrilateral ABDC below is a rhombus.  
What is the value of the angle  $x^\circ$  marked on the diagram?



(A)  $18^\circ$                       (B)  $36^\circ$                       (C)  $46^\circ$                       (D)  $72^\circ$

**End of Section I**

## Section II

90 marks

Attempt Questions 11 to 16

Allow about 2 hours 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available. In Questions 11-16, your responses should include relevant mathematical reasoning and/or calculations.

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Question 11 (15 marks)

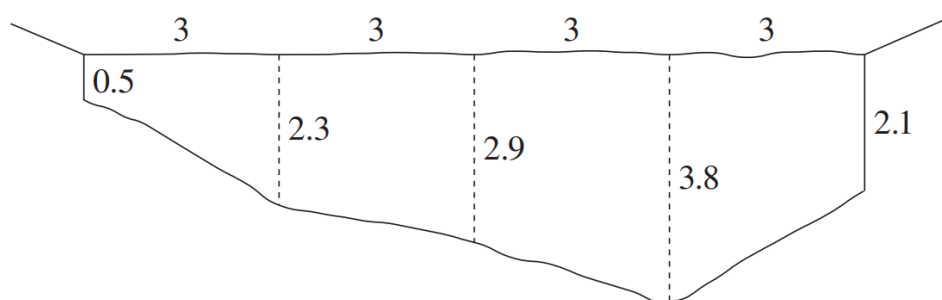
Start a new writing booklet.

- (a) Factorise  $9x^2 - 16$ . 1
- (b) Rationalise the denominator of  $\frac{1}{2 + \sqrt{5}}$ . 2
- (c) Solve  $|x - 1| \geq 4$ . 2
- (d) Differentiate  $(3 + e^{2x})^5$ . 2
- (e) Find  $\int \frac{6x^2}{x^3 + 1} dx$ . 2
- (f) Solve  $\sin^2 x + 2 \cos x = 1$  for  $0 \leq x \leq 2\pi$ . 3
- (g) Sketch the region defined by  $(x - 2)^2 + (y - 3)^2 < 4$ . 3

End of Question 11

**Question 12 (15 marks)****Start a new writing booklet.**

- (a) Solve  $5^x = 4$  correct to 1 decimal place. 2
- (b) Find the equation of the tangent to the curve  $y = e^x$  at the point  $(1, e)$ . 2
- (c) Evaluate the definite integral 2  
$$\int_{-1}^3 (6x - 7) dx.$$
- (d) The gradient of a curve is given by  $\frac{dy}{dx} = 9x^2 - 2x + 1$ . The curve passes through 3  
the point  $(-1, -4)$ . What is the equation of the curve?
- (e) A sum of \$20 000 is invested at a fixed rate of interest, compounded annually. 3  
After 5 years the principal has grown to \$28 567.  
Find the annual rate of interest as a percentage correct to one decimal place.
- (f) Below is a diagram showing the cross-section of a creek, with depths of the creek 2  
given in metres, at 3 metre intervals. The creek is 12 metres in width.

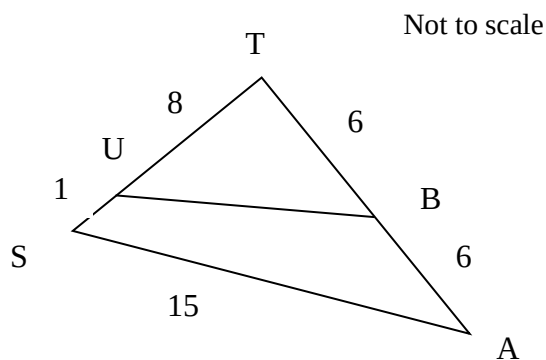


- (i) Use Simpson's rule with five depth measurements to calculate the 2  
approximate area of the cross-section.
- (ii) If the water flows through this section of the creek at  $0.5 \text{ m}^{-1}$ . Calculate the 1  
approximate volume of water that flows past this section in ten seconds.

**End of Question 12**

**Question 13 (15 marks)****Start a new writing booklet.**

- (a) Evaluate 2
- $$\int_0^{\frac{\pi}{2}} \sec^2 \frac{x}{2} dx.$$
- (b) (i) Show that the perpendicular distance from the point  $(1, -4)$  to the line  $4x - 3y + 14 = 0$  is 6 units. 1
- (ii) Find the centre and radius of the circle  $x^2 - 2x + y^2 + 8y = 8$ . 2
- (iii) Explain why the line  $4x - 3y + 14 = 0$  will never intersect the circle  $x^2 - 2x + y^2 + 8y = 8$ . 1
- (c) A particle is moving in a straight line. At time  $t$  seconds its displacement is  $x$  metres from the fixed point  $O$  on the line and its velocity is given by  $v = 3t^2 - 2t - 1$ . Initially the particle is 1 metre to the right of  $O$ .
- (i) Show that the particle is at rest after 1 second. 1
- (ii) Find the displacement  $x$  in terms of  $t$ . 2
- (iii) Find the distance travelled by the particle in the first 2 seconds. 2
- (d)

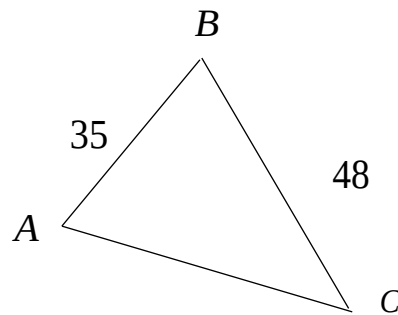


- (i) Prove that  $\triangle BUT$  is similar to  $\triangle SAT$ . 2
- (ii) Hence, or otherwise, find the length of  $BU$ . 2

**End of Question 13**

**Question 14 (15 marks)****Start a new writing booklet.**

- (a) The bearing of  $B$  from  $A$  is  $036^\circ T$  and the bearing of  $C$  from  $B$  is  $156^\circ T$ .



Not to scale

Copy the diagram into your writing booklet

- (i) Show that the value of  $\angle ABC = 60^\circ$ . 2
- (ii) Find the distance  $AC$ . 2
- (iii) Find the bearing of  $A$  from  $C$ . 2
- (b)
- (i) Sketch the curve  $y = 3 + \cos 2x$  for  $-\pi \leq x \leq \pi$ . 3
- (ii) Find the exact value of the area under the curve  $y = 3 + \cos 2x$  between the  $x$ -axis,  $x = 0$  and  $x = \frac{7\pi}{12}$ . 2
- (c) Two players, in a game, take turns at drawing and then immediately replacing a marble from a bag. The bag contains 2 green and 3 red marbles. Player  $A$  draws first.  
For  $A$  to win he must draw a green marble.  
For  $B$  to win he must draw a red marble.
- Find the probability that:
- (i)  $A$  wins on his first draw. 1
- (ii)  $B$  wins on his first draw. 1
- (ii)  $A$  wins in fewer than 4 of his turns. 2

**End of Question 14**

**Question 15 (15 marks)****Start a new writing booklet.**

- (a)
- (i) Differentiate  $x^2 \ln x$ . 1
- (ii) Hence, or otherwise, find  $\int 3x(1 + \ln x^2) dx$ . 2
- (b) Consider the function  $f(x) = 3 - 3x^2 - x^3$  in the domain  $-3 \leq x \leq 2$ .
- (i) Find the stationary points and determine their nature. 3
- (ii) Find the point of inflexion. 2
- (iii) Draw a sketch of the curve  $y = f(x)$  in the domain  $-3 \leq x \leq 2$ . 2
- (iv) What is the minimum value of the function in the given domain? 1
- (c) Jamie borrows \$35000 at 18% p.a. reducible interest. She plans to repay the loan in equal monthly instalments over 5 years. If  $A_n$  is the amount owing after  $n$  instalments and  $M$  is the amount paid in each instalment:
- (i) Show that  $A_2 = 35000 \times 1.015^2 - M(1.015 + 1)$  1
- (ii) Find the amount she needs to pay for each instalment to pay off the loan. Answer to the nearest dollar. 3

**End of Question 15**

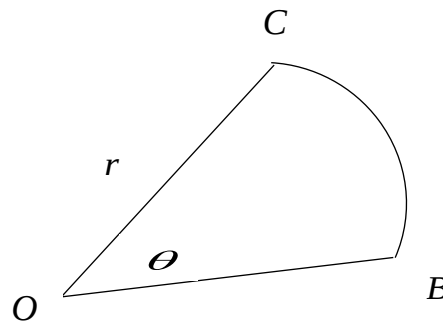
**Question 16 (15 marks)****Start a new writing booklet.**

- (a) Find the exact value of the volume of the solid formed when the area bounded by the curve  $y = 3 - x^2$ , for  $x \geq 0$ , the  $y$ -axis and the line  $y = 2$  is rotated about the  $x$ -axis. 4

- (b) The death rate of an endangered species on an island is given by
- $$\frac{dP}{dt} = -kP,$$
- where  $P$  is the population of the species after  $t$  days and  $k$  is a constant.

- (i) Show that  $P = Ae^{-kt}$  is a solution to the equation, where  $A$  is a constant. 1
- (ii) Initially there were 2000 of the species on the island, after 300 days only 1000 were left. What is the population (to the nearest whole number) after 400 days? 2
- (iii) After how many days will the population drop below 400? 2

- (c) The diagram below shows a sector of a circle with centre  $O$  and radius  $r$  cm. The arc subtends an angle  $\theta$  radians at  $O$  and the area of the sector is  $8 \text{ cm}^2$ .



- (i) Find an expression for  $r$  in terms of  $\theta$ . 1
- (ii) Show that the perimeter of the sector is given by  $P = \frac{8}{\sqrt{\theta}} + 4\sqrt{\theta}$ . 1
- (iii) If  $0 \leq \theta \leq \pi$ , find the value of  $\theta$  for a minimum perimeter. 4

**End of paper**



Student Number: \_\_\_\_\_

Teacher: \_\_\_\_\_

Class: \_\_\_\_\_

FORT STREET HIGH SCHOOL

## 2018

### HIGHER SCHOOL CERTIFICATE COURSE

## ASSESSMENT TASK 3: SOLUTIONS

# Mathematics

**Time allowed: 3 hours**

(plus 5 minutes reading time)

| Syllabus Outcomes | Assessment Area Description and Marking Guidelines                                                                                                                | Questions  |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
|                   | Chooses and applies appropriate mathematical techniques in order to solve problems effectively                                                                    | 1-10       |
| H2, H3, H4, H5    | Manipulates algebraic expressions to solve problems from topic areas such as geometry, co-ordinate geometry, quadratics, trigonometry, probability and logarithms | 12, 14     |
| H6, H7, H8        | Demonstrates skills in the processes of differential and integral calculus and applies them appropriately                                                         | 11, 13, 15 |
| H9                | Synthesises mathematical solutions to harder problems and communicates them in appropriate form                                                                   | 16         |

### Total Marks 100

#### Section I 10 marks

Multiple Choice, attempt all questions,  
Allow about 15 minutes for this section

#### Section II 90 Marks

Attempt Questions 11-16,  
Allow about 2 hours 45 minutes for this section

#### General Instructions:

- Questions 11-16 are to be started in a new booklet
- The marks allocated for each question are indicated
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations.
- Marks may be deducted for careless or badly arranged work.
- Board – approved calculators may be used

| Section I  | Total 10 | Marks |
|------------|----------|-------|
| Q1-Q10     |          |       |
| Section II | Total 90 | Marks |
| Q11        | /15      |       |
| Q12        | /15      |       |
| Q13        | /15      |       |
| Q14        | /15      |       |
| Q15        | /15      |       |
| Q16        | /15      |       |
|            | Percent  |       |

# Section I

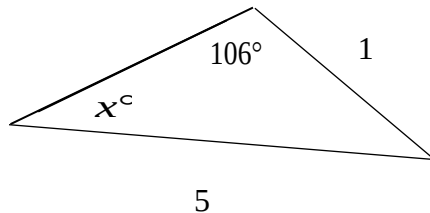
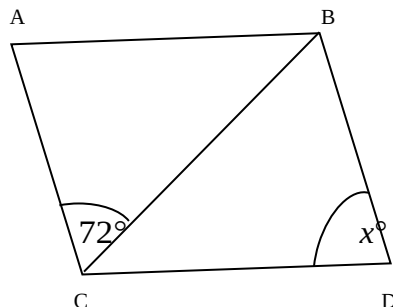
10 marks

Attempt Questions 1 to 10

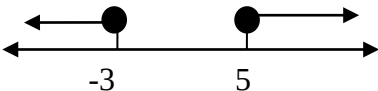
Allow about 15 minutes for this section

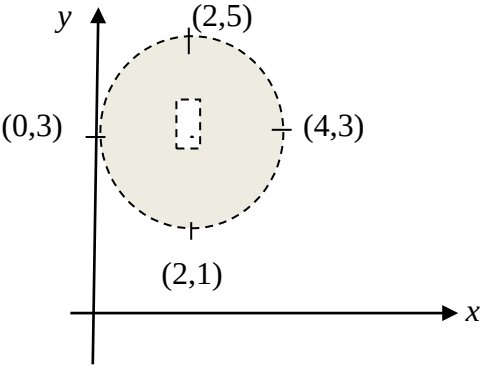
Use the multiple-choice answer sheet for Questions 1 to 10

|    |                                                                                                                             |                                           |                                                     |                                                                         |
|----|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|-----------------------------------------------------|-------------------------------------------------------------------------|
| 1. | What is the locus of a set of points that is equidistant from a fixed point and a fixed line?                               |                                           |                                                     |                                                                         |
|    | (A) <u>a parabola</u>                                                                                                       | (B) a hyperbola                           | (C) a circle                                        | (D) a straight line                                                     |
| 2. | Which of the following quadratic equations have two distinct real roots?                                                    |                                           |                                                     |                                                                         |
|    | (A) $y = x^2 - 4x + 4$                                                                                                      | (B) $y = x^2 + 4x + 4$                    |                                                     |                                                                         |
|    | (C) <u><math>y = x^2 - 4x - 4</math></u>                                                                                    | (D) $y = x^2 + 4$                         |                                                     |                                                                         |
| 3. | The solutions of $\sqrt{3} \tan x = -1$ for $0 \leq x \leq 2\pi$ are?                                                       |                                           |                                                     |                                                                         |
|    | (A) $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$                                                                                   | (B) $\frac{2\pi}{3}$ and $\frac{5\pi}{3}$ | (C) $\frac{5\pi}{6}$ and $\frac{7\pi}{6}$           | (D) <u><math>\frac{5\pi}{6}</math> and <math>\frac{11\pi}{6}</math></u> |
| 4. | Find the limiting sum of the geometric series $\frac{2}{3} - \frac{2}{15} + \frac{2}{75} - \frac{2}{375} + \dots$           |                                           |                                                     |                                                                         |
|    | (A) $\frac{3}{5}$                                                                                                           | (B) 0                                     | (C) $\frac{12}{15}$                                 | (D) <u><math>\frac{5}{9}</math></u>                                     |
| 5. | Which of the following conditions for $\frac{dP}{dt}$ and $\frac{d^2P}{dt^2}$ describes the slowing growth of a variable B? |                                           |                                                     |                                                                         |
|    | (A) $\frac{dP}{dt} > 0$ and $\frac{d^2P}{dt^2} > 0$                                                                         |                                           | (B) $\frac{dP}{dt} < 0$ and $\frac{d^2P}{dt^2} < 0$ |                                                                         |
|    | (C) <u><math>\frac{dP}{dt} &gt; 0</math></u> <u>and</u> <u><math>\frac{d^2P}{dt^2} \leq 0</math></u>                        |                                           | (D) $\frac{dP}{dt} < 0$ and $\frac{d^2P}{dt^2} > 0$ |                                                                         |

|     |                                                                                                                                                                                                               |                                  |                                                          |                                     |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|----------------------------------------------------------|-------------------------------------|
| 6.  | The quadratic equation $x^2 + 4x - 1 = 0$ has roots $\alpha$ and $\beta$ .<br>What is the value of $\alpha\beta + (\alpha + \beta)$ ?                                                                         |                                  |                                                          |                                     |
|     | (A) 5                                                                                                                                                                                                         | (B) 3                            | (C) <u>-5</u>                                            | (D) -3                              |
| 7.  | If $\ln a = \ln b + \ln c$ , then which of these is true?                                                                                                                                                     |                                  |                                                          |                                     |
|     | (A) <u><math>a = bc</math></u>                                                                                                                                                                                | (B) $a = b + c$                  | (C) $\ln a = bc$                                         | (D) $a = \frac{b}{c}$               |
| 8.  | Which calculation gives the value of $X$ in the diagram below? <div></div>                                                   |                                  |                                                          |                                     |
|     | (A) <u><math>x = \sin^{-1}\left(\frac{\sin 106^\circ}{5}\right)</math></u>                                                                                                                                    |                                  | (B) $x = \cos^{-1}\left(\frac{\sin 106^\circ}{5}\right)$ |                                     |
|     | (C) $x = \sin^{-1}(5\sin 106^\circ)$                                                                                                                                                                          |                                  | (D) $x = \sin^{-1}\left(\frac{1}{5}\right)$              |                                     |
| 9.  | A bag contains red and green lollies in the ratio of 7 : 2 . If a lolly is selected at random, what is the probability that it is a green lolly?                                                              |                                  |                                                          |                                     |
|     | (A) $\frac{1}{7}$                                                                                                                                                                                             | (B) $\frac{7}{9}$                | (C) $\frac{2}{7}$                                        | (D) <u><math>\frac{2}{9}</math></u> |
| 10. | The quadrilateral ABDC below is a rhombus.<br>What is the value of the angle $x^\circ$ marked on the diagram? <div></div> |                                  |                                                          |                                     |
|     | (A) $18^\circ$                                                                                                                                                                                                | (B) <u><math>36^\circ</math></u> | (C) $46^\circ$                                           | (D) $72^\circ$                      |

### End of Section I

|                                      |                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                          |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Section II</b><br><b>90 marks</b> |                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                          |
| <b>Question 11 (15 marks)</b>        |                                                                                                                                                                                                                                                                                                                           | <b>Teacher's Comments</b>                                                                                                                                                                                                                                |
| <b>(a)</b>                           | $9x^2 - 16 = (3x)^2 - (4)^2$ $= (3x - 4)(3x + 4) \quad \text{①}$                                                                                                                                                                                                                                                          | <i>Well Done</i>                                                                                                                                                                                                                                         |
| <b>(b)</b>                           | $\frac{1}{2 + \sqrt{5}} \times \frac{2 - \sqrt{5}}{2 - \sqrt{5}} = \frac{2 - \sqrt{5}}{4 - 5} \quad \text{①}$ $= \frac{2 - \sqrt{5}}{-1}$ $= \sqrt{5} - 2 \quad \text{①}$                                                                                                                                                 | <i>Well done</i><br><i>A few students did not simplify fully</i>                                                                                                                                                                                         |
| <b>(c)</b>                           | $ x - 1  \geq 4$<br>$x - 1 \geq 4, x - 1 \leq -4$<br>$x \geq 5, \text{①} \quad x \leq -3 \quad \text{①}$<br>                                                                                                                             | <i>Mostly Well done</i><br><i>Some students forgetting to reverse inequality sign when dividing/ multiplying by a negative</i><br><br><i>Some answers showing students obviously did not test their results</i><br><br><i>-3 ≥ x ≥ 5 is unacceptable</i> |
| <b>(d)</b>                           | $\frac{d}{dx} (3 + e^{2x})^5 = 5(3 + e^{2x})^4 \cdot e^{2x} \cdot 2 \quad \text{①}$ $= 10e^{2x} (3 + e^{2x})^4 \quad \text{①}$                                                                                                                                                                                            | <i>Well done</i><br><br><i>Very few students failing to simplify the answer, need to recognise the mark allocated, requiring a simplified answer</i>                                                                                                     |
| <b>(e)</b>                           | $\int \frac{6x^2}{x^3 + 1} dx = 2 \int \frac{3x^2}{x^3 + 1} dx$ $= 2 \ln(x^3 + 1) + c$ <p style="text-align: center;">①                      ①</p>                                                                                                                                                                        | <i>Most students recognised the integral as a log.</i><br><br><i>Care needed not to leave off the brackets and/ or the constant</i>                                                                                                                      |
| <b>(f)</b>                           | $\sin^2 x + 2 \cos x = 1$ $1 - \cos^2 x + 2 \cos x - 1 = 0 \quad \text{Let } \cos x = u$ $1 - u^2 + 2u - 1 = 0$ $-u^2 + 2u = 0 \quad \text{①}$ $u(2 - u) = 0$ $\therefore u = 0, u = 2$ $\cos x = 0, \quad \cos x = 2 \text{ (no solution)} \quad \text{①}$ $\therefore x = \frac{\pi}{2}, \frac{3\pi}{2} \quad \text{①}$ | <i>Some students did not recognise the use of Trig. Identities.</i><br><br><i>Marks lost for not answering the question i.e. including solutions outside <math>0 \leq x \leq 2\pi</math></i>                                                             |

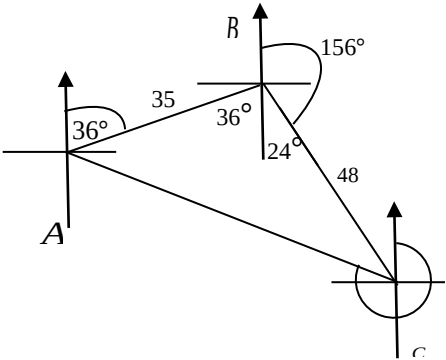
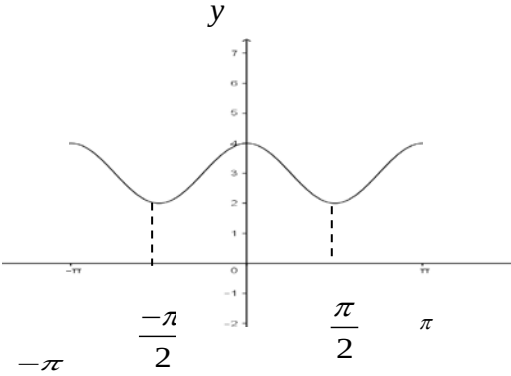
|                                                              |                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>(g)</b></p>                                            | <p><math>(x-2)^2 + (y-3)^2 &lt; 4</math></p> <p>Region Shaded inside circle ❶</p> <p>Broken line ❶</p> <p>Centre (2, 3), radius 2 ❶</p>  | <p>Many answers failed to show a broken line to exclude the points on the circle in region shaded.</p> <p>Many answers did not clearly label to show the features of the graph such as points to indicate the centre and radius of the circle.</p> <p>Some students did not recognise the equation as the graph of a circle.</p> |
| <p style="text-align: center;"><b>End of Question 11</b></p> |                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                  |

| Question 12 (15 marks) |                                                                                                                                                                                                                                                       | Teacher's Comments                                                                                                                                                                                                                |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>(a)</b></p>      | <p><math>5^x = 4</math></p> <p><math>\ln 5^x = \ln 4</math> ❶</p> <p><math>x \ln 5 = \ln 4</math></p> <p><math>x = \frac{\ln 4}{\ln 5}</math></p> <p><math>= 0.86135</math></p> <p><math>\approx 0.9</math> ❶</p> <p>(correct to 1 decimal place)</p> | <p><b>1 mark off for answering to 2dp (or other) instead of 1dp</b></p>                                                                                                                                                           |
| <p><b>(b)</b></p>      | <p><math>\frac{dy}{dx} = e^x</math> at <math>x=1</math></p> <p><math>m = e^x = e^1 = e</math> ❶</p> <p>Eqn. of tangent at (1, e)</p> <p><math>y - e = e(x - 1)</math></p> <p><math>y = ex - e + e</math></p> <p><math>y = ex</math> ❶</p>             | <p><b>Good use of point gradient formula but</b><br/> <math>m \neq e^x</math></p> <p>Sub <math>x=1</math>; ie:<br/> <math>m = e^x = e^1 = e</math></p> <p><b>General form is acceptable too:</b><br/> <math>ex - y = 0</math></p> |
| <p><b>(c)</b></p>      | <p><math>\int_{-1}^3 (6x - 7) dx = 3x^2 - 7x \Big _{-1}^3</math> ❶</p> <p><math>= [3(3)^2 - 7(3)] - [3(-1)^2 - 7(-1)]</math></p> <p><math>= [6] - [10]</math></p> <p><math>= -4</math> ❶</p>                                                          | <p><b>Watch your substitutions with negatives of negatives. Many people got +10 instead of -4 due to substitution errors.</b></p>                                                                                                 |

|                    |                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                            |
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|                    |                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                            |
| (d)                | $\int (9x^2 - 2x + 1) dx = \frac{9x^3}{3} - \frac{2x^2}{2} + x + c$ $\therefore y = 3x^3 - x^2 + x + c \quad \text{①}$ <p>since <math>(-1, -4)</math> lies on the curve then</p> $y = 3(-1)^3 - (-1)^2 + (-1) + c$ $-4 = -3 - 1 - 1 + c \quad \text{①}$ $\therefore c = 1 \text{ and } y = 3x^3 - x^2 + x + 1 \quad \text{①}$             | <p><b>Don't use the point gradient formula here. The point gradient formula is used for finding equations of straight lines, which are curves with constant gradients. This curve doesn't have a constant gradient, it's gradient is dependent on x.</b></p>                                                                                                                                                               |
| (e)                | $A = 20000 \left(1 + \frac{r}{100}\right)^5 \quad \text{①}$ $28567 = 20000 \left(1 + \frac{r}{100}\right)^5$ $\left(\sqrt[5]{\frac{28567}{20000}} - 1\right) \times 100 = r \quad \text{①}$ $\therefore r = 7.3907...$ $r \approx 7.4\% \quad \text{①}$                                                                                   | <p><b>Use compound interest, not simple interest.</b></p> <p><b>Convert your answer to a percentage, don't leave it as 0.074 or round to 0.1.</b></p> <p><b>Answer to 1dp as specified (not 2dp).</b></p>                                                                                                                                                                                                                  |
| (f)                | <p>(i) Area</p> $\approx \frac{h}{3} (y_0 + 4y_1 + 2y_2 + 4y_3 + 2y_4 + y_5)$ $\approx \frac{3}{3} [0.5 + 2.1 + 4(2.3 + 3.8) + 2(2.9)] \quad \text{①}$ $\approx 32.8 \text{ m}^2 \quad \text{①}$ <p>(ii) Distance = speed x time</p> $= 0.5 \times 10$ $= 5 \text{ m}$ $\text{Volume} = 32.8 \times 5$ $= 164 \text{ m}^3 \quad \text{①}$ | <p><b>Students made mistakes calculating h, which is the "strip" width, or smallest interval width, in this case: 3.</b></p> <p><b>It is also equal to (b – a) divided by the number of "strips" or smallest intervals, in this case: 12 / 4.</b></p> <p><b>ii) 164 m<sup>3</sup> not 5 and not 164L.</b><br/> <b>In maths, volume is m<sup>3</sup> and capacity is L.</b></p> <p><b>P.S.: 1 m<sup>3</sup> = 1000L</b></p> |
| End of Question 12 |                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                            |

| Question 13 (15 marks) |                                                                                                                                                                                                                                                                                      | Teacher's Comments                                                                                                                                                                                                             |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a)                    | $\int_0^{\frac{\pi}{2}} \sec^2 \frac{x}{2} dx = \left[ \frac{\tan \frac{x}{2}}{\frac{1}{2}} \right]_0^{\frac{\pi}{2}}$ $= 2 \tan \frac{x}{2} \Big _0^{\frac{\pi}{2}} \quad \text{①}$ $= 2 \left[ \tan \frac{\pi}{4} - \tan 0 \right]$ $= 2[1 - 0]$ $= 2 \quad \text{①}$              | <p>Generally well done.</p> <p>Some students placing the limits at the start of the brackets – e.g. <math>\frac{\pi}{2} \left[ 2 \tan \frac{x}{2} \right]_0</math> which is not correct.</p>                                   |
| (b)                    | <p>(i)</p> $d = \frac{ 4(1) - 3(-4) + 14 }{\sqrt{4^2 + 3^2}} \quad \text{①}$ $= \frac{ 4 + 12 + 14 }{\sqrt{25}}$ $= \frac{ 30 }{5}$ $= 6$                                                                                                                                            | Generally well done.                                                                                                                                                                                                           |
|                        | <p>(ii)</p> $x^2 - 2x + y^2 + 8y = 8$ $x^2 - 2x + 1 + y^2 + 8y + 16 = 8 + 1 + 16.$ $(x - 1)^2 + (y + 4)^2 = 25 \quad \text{①}$ <p>Giving a circle centre (1, -4), radius 5      ①</p>                                                                                                | Some students had trouble completing the square correctly.                                                                                                                                                                     |
|                        | <p>(iii) The line <math>4x - 3y + 14 = 0</math> will never intersect the circle <math>x^2 - 2x + y^2 + 8y = 8</math> since the distance of the line from the point (which is the centre of the circle) is 6 units and this is greater than the circles radius of 5 units.      ①</p> | Many students did not use the previous parts to easily answer this one. Only some were able to show $\Delta < 0$ to give the no intercept result. A few tried graphing, but without grid paper, accuracy cannot be guaranteed. |
| (c)                    | <p>(i) The particle is at rest (stationary) when <math>v = 0</math><br/>When <math>t = 1</math>,</p> $v = 3(1)^2 - 2(1) - 1$ $= 3 - 2 - 1$ $= 0$ <p><math>\therefore</math> the particle is at rest after 1sec      ①</p>                                                            | Generally well done.                                                                                                                                                                                                           |

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| (c)                | <p>(ii)</p> $x = \int (3t^2 - 2t - 1) dt$ $= \frac{3}{3}t^3 - \frac{2}{2}t^2 - t + c$ $= t^3 - t^2 - t + c \quad \text{①}$ <p>When <math>t = 0</math>, <math>x = 1</math></p> $\therefore 1 = 0^3 - 0^2 - 0 + c$ $c = 1$ <p>thus <math>x = t^3 - t^2 - t + 1 \quad \text{①}</math></p> <p>(displacement <math>x</math> in terms of <math>t</math>)</p>                                                                                                                     | <p>Generally well done, but many abbreviated too many steps or made careless transcription errors.</p>                                                                                                                                                                                                                                                                                                                                     |
|                    | <p>(iii)</p> $x = \left  \int_0^1 (3t^2 + -2t - 1) dt \right  + \left  \int_1^2 (3t^2 + -2t - 1) dt \right $ $= \left  \left[ (t^3 - t^2 - 1) \right]_0^1 \right  + \left  \left[ (t^3 - t^2 - 1) \right]_1^2 \right  \quad \text{①}$ $= \left  \left[ (1 - 1 - 1) - 0 \right] \right  + \left  \left[ (8 - 4 - 2) - (1 - 1 - 1) \right] \right $ $=  -1  +  3 $ $= 4 \quad \text{①}$ <p><math>\therefore</math> the particle travels 4 metres in the first 2 seconds.</p> | <p>Many students did not realize the significance of the stat. pt. at <math>x = 1</math> from the previous parts.</p> <p><b>Note:</b> displacement <math>\neq</math> distance travelled!!</p> <p>The most successful alternate solution was summing the distance travelled between the 0 and 1 second marks with that travelled between the 1 and 2 second marks.</p>                                                                      |
| (d)                | <p>(i)</p> <p><math>\angle T</math> is common <span style="float: right;">①</span></p> <p>From <math>\triangle BUT : \triangle SAT</math></p> $\frac{UT}{AT} = \frac{8}{12} = \frac{2}{3} \text{ and } \frac{BT}{ST} = \frac{6}{9} = \frac{2}{3} \quad \text{①}$ <p><math>\triangle BUT</math> is similar to <math>\triangle SAT</math></p> <p>(two corresponding sides are in the same ratio and the included angle is equal)</p>                                         | <p>Many students started by stating that the ratios were equal (as if this was given), rather than showing that they were equal.</p> <p>i.e. <math>\frac{UT}{AT} = \frac{BT}{ST} = \frac{2}{3}</math>. <b>This was not given any marks.</b></p> <p>Some students also made no mention of an angle, as if doing a SSS in ratio proof, but as one side is unknown, this is not possible. An angle is needed in the proof. (SAS in ratio)</p> |
|                    | <p>(ii)</p> <p><math>BU = 10</math> as <math>\triangle BUT</math> is a right <math>\triangle</math> <span style="float: right;">①</span></p> <p>(6,8,10 pythag.triad) <span style="float: right;">①</span></p> <p>OR</p> $\frac{BU}{15} = \frac{2}{3} \therefore BU = \frac{30}{3} = 10 \quad \text{①}$ <p>(corresponding sides of similar <math>\triangle s</math>) <span style="float: right;">①</span></p> <p>(are proportional)</p>                                    | <p>Generally well done.</p>                                                                                                                                                                                                                                                                                                                                                                                                                |
| End of Question 13 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| Question 14 (15 marks)                                                                                                                                                                                                             | Teacher's Comments |
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| <p>(a)</p>  <p>❶ diagram OR reason (see below )</p>                                                                                               |                    |
| <p>(i)</p> <p><math>\angle ABC = 36^\circ + 24^\circ</math> (alternate <math>\angle</math>'s on parallel lines, adj. sup. <math>\angle</math>'s)</p> <p><math>= 60^\circ</math>    ❶</p>                                           |                    |
| <p>(ii)</p> <p><math>d^2 = 35^2 + 48^2 - 2 \times 35 \times 48 \times \cos 60^\circ</math>    ❶</p> <p><math>= 1849</math></p> <p><math>d = \sqrt{1849}</math></p> <p><math>= 43</math>    ❶</p> <p>the distance of AC is 43km</p> |                    |
| <p>(iii)</p> <p>The bearing of A from C</p> <p><math>= 360^\circ - (\angle BCA + 24^\circ)</math>    ❶</p> <p><math>= 360 - (44^\circ 49' + 24^\circ)</math></p> <p><math>= 291^\circ 11'</math>    ❶</p>                          |                    |
| <p>(b) (i)</p> <p><math>y = 3 + \cos 2x</math> for <math>-\pi \leq x \leq \pi</math></p>  <p>❶ shape    ❶ y-intercept    ❶ endpoints</p>        |                    |

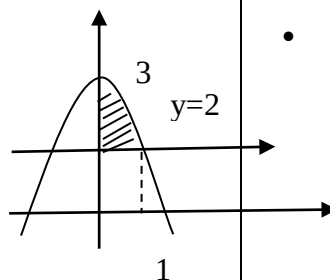
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|                    |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |
|                    |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |
|                    | (ii) | $A = \int_0^{\frac{7\pi}{12}} (3 + \cos 2x) dx$ $= \left[ 3x + \frac{\sin 2x}{2} \right]_0^{\frac{7\pi}{12}} \quad \text{①}$ $= \left[ 3\left(\frac{7\pi}{12}\right) + \frac{1}{2}\left(\sin \frac{7\pi}{6}\right) \right] - \left[ 3(0) + \frac{1}{2}(\sin 0) \right]$ $= \left[ \left(\frac{7\pi}{4}\right) + \frac{1}{2}\left(\sin \frac{7\pi}{6}\right) \right] - [0]$ $= \frac{7\pi}{4} - \frac{1}{4}$ $= \frac{7\pi - 1}{4} \text{ units}^2 \quad \text{①}$ |  |
| (c)                |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |
|                    | (i)  | $P(A \text{ wins in one turn}) = \frac{2}{5} \quad \text{①}$                                                                                                                                                                                                                                                                                                                                                                                                      |  |
|                    | (ii) | $B \text{ (wins on his first draw)}$ $= \frac{3}{5} \times \frac{3}{5}$ $= \frac{9}{25} \quad \text{①}$                                                                                                                                                                                                                                                                                                                                                           |  |
|                    |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |
|                    | (ii) | $P(A \text{ wins in fewer than 4 turns})$ $= P(A \text{ wins in 1 turn}) + P(A \text{ wins on 2nd turn}) + P(A \text{ wins in 3 turns})$ $= \frac{2}{5} + \left(\frac{3}{5} \times \frac{2}{5} \times \frac{2}{5}\right) + \left(\frac{3}{5} \times \frac{2}{5} \times \frac{3}{5} \times \frac{2}{5} \times \frac{2}{5}\right) \quad \text{①}$ $= \frac{2}{5} + \frac{12}{125} + \frac{72}{3125}$ $= \frac{1622}{3125} \quad \text{①}$                           |  |
| End of Question 14 |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |

| Question 15 (15 marks) |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Teacher's Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a)                    |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                        | (i)  | $\frac{d}{dx}(x^2 \ln x) = x^2 \frac{d}{dx}(\ln x) + \ln x \frac{d}{dx}(x^2)$ $= x^2 \left(\frac{1}{x}\right) + (2x) \ln x$ $= x + 2x \ln x \quad \text{①}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Generally well done.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                        | (ii) | $\int 3x(1 + \ln x^2) dx$ $= 3 \int x + x \ln x^2 dx$ $= 3 \int x + 2x \ln x dx \quad \text{① (as } \ln x^n = n \ln x)$ $= 3(x^2 \ln x) + c \quad \text{①}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>Many were unable to use the log laws and realise <math>\ln x^2 \equiv 2 \ln x</math>.</p> <p>Many fudges in answer around this point.</p>                                                                                                                                                                                                                                                                                                                                                                                                  |
| (b)                    |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                        | (i)  | $f(x) = 3 - 3x^2 - x^3 \text{ in the domain } -3 \leq x \leq 2$ $f'(x) = -6x - 3x^2$ $f''(x) = -6 - 6x$ <p><math>\therefore</math> Stationary points occur when <math>f'(x) = 0</math></p> $f'(x) = -6x - 3x^2$ <p>i.e. <math>-6x - 3x^2 = 0</math></p> $-3x(2 + x) = 0$ <p><math>\therefore x = 0, x = -2 \quad \text{①}</math></p> <p>Nature of Stationary points:</p> <p>When</p> $x = 0, y = 3 \text{ and } f''(x) = -6 - 6(0)$ $< 0 \text{ (concave down)}$ <p><math>\therefore (0, 3)</math> is a maximum turning point <math>\text{①}</math></p> <p>When</p> $x = -2, y = 3 - 3(-1)^2 - (-2)^3 = -1$ <p>and <math>f''(x) = -6 - 6(-2)</math></p> $> 0 \text{ (concave up)}$ <p><math>\therefore (-2, -1)</math> is a minimum turning point <math>\text{①}</math></p> | <p>Many did not take note the domain required, particularly for the sketch later.</p> <p>Some students are unclear on the following:</p> <p><math>f'(x) = 0</math> always produces a stat. pt. You must always test for the <u>type</u> of stat pt.</p> <p><math>f''(x) = 0</math> only produces a possible inflection point. You must always test for concavity change for a point of inflection.</p> <p>A few did not state points as required, but just gave <math>x</math> and <math>y</math> values. Otherwise, generally well done.</p> |
|                        | (ii) | Points of inflexion occur when $f''(x) = 0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                   |          |          |         |          |       |       |       |                                   |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|---------|----------|-------|-------|-------|-----------------------------------|
|          | $f''(x) = -6 - 6x$<br><i>i.e.</i> $-6 - 6x = 0$<br>$-6(1+x) = 0$<br>$\therefore x = -1$ and $y = 3 - (-1)^2 - (-1)^3$<br>$= 1$<br>possible point of inflexion at $(-1, 1)$ ❶<br><br>(ii)cont.<br><table><tr><td><math>x</math></td><td><math>x = -2</math></td><td><math>x = -1</math></td><td><math>x = 0</math></td></tr><tr><td><math>f''(x)</math></td><td><math>&gt; 0</math></td><td><math>= 0</math></td><td><math>&lt; 0</math></td></tr></table><br>Sign/ concavity change either side<br>$\therefore$ point of inflexion at $(-1, 1)$ ❶ | $x$                                                                                                                                                                                                               | $x = -2$ | $x = -1$ | $x = 0$ | $f''(x)$ | $> 0$ | $= 0$ | $< 0$ | Many did not test for inflection. |
| $x$      | $x = -2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $x = -1$                                                                                                                                                                                                          | $x = 0$  |          |         |          |       |       |       |                                   |
| $f''(x)$ | $> 0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $= 0$                                                                                                                                                                                                             | $< 0$    |          |         |          |       |       |       |                                   |
|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                   |          |          |         |          |       |       |       |                                   |
| (iii)    | <br><br>❶ shape & pt of inflexion<br>❶ y intercept & endpoints                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | End-points of domain frequently missed – these must be there when a domain is given!<br><br>A sketch needs to be in scale, be neat, in pencil and showing the points found. Many diagrams were very poor quality. |          |          |         |          |       |       |       |                                   |
| (iv)     | Minimum value of the function is -17 ❶                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Minimum value of a function is a number - <u>not</u> a point or a y-value!                                                                                                                                        |          |          |         |          |       |       |       |                                   |
|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                   |          |          |         |          |       |       |       |                                   |
| (c)      | $A = \$35000$<br>$n = 5 \times 12 = 60$ months<br>$r = \frac{18}{12}\% = 1.5\% = 0.015$ per month                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                   |          |          |         |          |       |       |       |                                   |
| (i)      | Let $A_n$ be the amount owing after $n$ payments<br>$M$ be the amount paid monthly/each instalment                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                   |          |          |         |          |       |       |       |                                   |

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                   |
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|                    | $\therefore A_1 = 35000 + 35000 \times 0.015 - M$ $= 35000(1 + 0.015) - M$ $A_2 = A_1 \times 1.015 - M$ $= [35000(1.015) - M] \times 0.015 - M \text{ ❶}$ $= 35000 \times 1.015^2 - M(1.015) - M$ $= 35000 \times 1.015^2 - M(1.015 + 1)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p><i>Most successfully established the modification to <math>A_1</math> to show <math>A_2</math>. Those who lost this mark did not make the link to <math>A_1</math>.</i></p>                                                                                                                                                                                    |
| (ii)               | <p>Continued from part (i)</p> $A_3 = A_2(1.015) - M$ $= [35000(1.015)^2 - M(1.015 + 1)] \times 0.015 - M$ $= 35000(1.015)^3 - M[(1.015)^2 + 1.015] - M$ $= 35000(1.015)^3 - M(1.015^2 + 1.015 + 1)$ $\vdots$ $A_{60} = A_{59}(0.015) - M$ $= 35000 \times 1.015^{60} - M(1.015^{59} + 1.015^{58} + 1.015^{57} + \dots + 1) \text{ ❶}$ <p>(as a geometric series with <math>a=1</math>, <math>r=1.015</math>, <math>n=60</math>, gives) ❶</p> $= 35000 \times 1.015^{60} - M \left( \frac{1 \times (1.015^{60} - 1)}{1.015 - 1} \right)$ <p><math>A_{60} = 0</math> when loan is repaid after 60 months:</p> $0 = 35000 \times 1.015^{60} - M \left( \frac{1 \times (1.015^{60} - 1)}{1.015 - 1} \right)$ $M = \frac{(35000 \times 1.015^{60}) \times 0.015}{1.015^{60} - 1}$ $= 888.769$ $= \$889 \text{ (to the nearest dollar) } \text{ ❶}$ | <p><i>Most were also less successful in establishing the continuing pattern and generalizing for <math>A_n</math></i></p> <p><i>Many also did not make the link with a GP – make sure you state the values of <math>a, r</math> and <math>n</math> that apply.</i></p> <p><i>Most got the correct answer, although some poor calculator work was evident.</i></p> |
| End of Question 15 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                   |

| Question 16 (15 marks) |     |                                                                                                                                                                                                                                                                                                                                                                                                                                        | Teacher's Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| (a)                    |     | $y = 3 - x^2 \therefore y^2 = 9 - 6x^2 + x^4$ <p>when <math>y = 2, x = 1</math> ❶</p> $V = \pi \int_0^1 (y^2) dx - \pi \int_0^1 (2^2) dx$ $\therefore V = \pi \int_0^1 (9 - 6x^2 + x^4) dx - \pi \int_0^1 (4) dx$ ❶ $= \pi \left[ 9x - \frac{6x^3}{3} + \frac{x^5}{5} \right]_0^1 - (4x)_0^1$ ❶ $= \pi \left[ 9(1) - 2(1)^3 + \frac{(1)^5}{5} \right] - 4(1) - [0]$ $= \pi \left[ \frac{36 - 20}{5} \right]$ $= \frac{16\pi}{5} u^3$ ❶ | <ul style="list-style-type: none"> <li>Not answered very well</li> <li>Students calculated the area bounded by the <math>x</math>-axis, the curve, the <math>y</math>-axis and the line <math>y=2</math> and did not realise that the question did not refer to the <math>x</math>-axis as a boundary. This led to them calculating the area under the line <math>y=2</math> and using incorrect <math>x</math> limits of 0 and <math>\sqrt{3}</math></li> <li>A few students rotated the curve about the <math>y</math>-axis instead of the <math>x</math>-axis</li> <li>There were quite a few careless errors such as forgetting to subtract area under the line when calculating the volume.</li> <li>Conceptual errors were also apparent with some students using <math>(3 - x^2 - 2)^2</math> in their calculations, indicating the need to improve algebraic skills</li> </ul> |
| (b)                    | (i) | $P = Ae^{-kt}$ $\frac{dP}{dt} = -k Ae^{-kt}$ $= -k P \quad (\text{as } P = Ae^{-kt})$ $\therefore P = Ae^{-kt} \text{ is a solution}$ ❶                                                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>Generally answered well</li> <li>A few students tried to use integration but were unable to arrive at the correct answer</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |



|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                            |
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|  | <p><b>(ii)</b> Given <math>t = 0, P = 2000</math> and <math>t = 300, P = 1000</math><br/> <math>\therefore</math> when <math>t = 0, P = Ae^{-kt}</math><br/> <math>2000 = Ae^0</math><br/> <math>A = 2000</math><br/> <math>t = 300, P = 1000</math><br/> <math>1000 = 2000(e^{-k(300)})</math><br/> <math>\ln \frac{1}{2} = -300k</math><br/> <math>\therefore k = \frac{\ln \frac{1}{2}}{-300}</math><br/> <math>= \frac{\ln 2}{300}</math> (since <math>\ln \frac{1}{2} = \ln 2^{-1} = -\ln 2</math>)<br/> <math>= 0.00231</math> ❶<br/> When <math>t = 400, P = 2000e^{-k(400)}</math><br/> <math>= 793.7</math><br/> <math>\approx 794</math> ❶<br/> After 400 days the population of the species is 794 (nearest whole number)</p> | <ul style="list-style-type: none"> <li>• Generally answered well</li> <li>• Students should round up in their final answer and use the exact value of <math>k</math> in subsequent calculations</li> </ul> |
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                            |
|  | <p><b>(iii)</b> <math>400 = 2000e^{-kt}</math> ❶<br/> <math>\ln \frac{400}{2000} = \ln e^{-kt}</math><br/> <math>\ln \frac{1}{5} = -kt</math><br/> <math>t = \frac{\ln \frac{1}{5}}{-k}</math><br/> <math>= 696.6</math><br/> <math>\therefore</math> after 697 days population of the species will drop below 400 ❶</p>                                                                                                                                                                                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>• Generally answered well</li> <li>• Students were awarded marks if correct calculations performed with a carry on error</li> </ul>                                 |
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                            |

|     |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (c) | (i)   | $\text{Area} = \frac{1}{2} r^2 \theta$ $8 = \frac{1}{2} r^2 \theta$ $\frac{16}{\theta} = r^2$ $r = \sqrt{\frac{16}{\theta}}, \quad r > 0$ $= \frac{4}{\sqrt{\theta}} \times \frac{\sqrt{\theta}}{\sqrt{\theta}}$ $= \frac{4\sqrt{\theta}}{\theta} \quad \text{①}$                                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>Generally answered well</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|     | (ii)  | $P = l_{BC} + 2r$ $= r\theta + 2\left(\frac{4}{\sqrt{\theta}}\right)$ $= \left(\frac{4}{\sqrt{\theta}}\right) \times \theta + 2\left(\frac{4}{\sqrt{\theta}}\right)$ $= \left(\frac{4\sqrt{\theta}}{\theta}\right) \times \theta + \frac{8}{\sqrt{\theta}} \quad \text{①}$ $P = \frac{8}{\sqrt{\theta}} + 4\sqrt{\theta}$                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>Generally answered well</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|     | (iii) | $P = \frac{8}{\sqrt{\theta}} + 4\sqrt{\theta} = 8\theta^{-\frac{1}{2}} + 4\theta^{\frac{1}{2}}$ $\therefore P' = -4\theta^{-\frac{3}{2}} + 2\theta^{-\frac{1}{2}}$ $= \frac{-4}{\sqrt{\theta^3}} + \frac{2}{\sqrt{\theta}}$ $= \frac{-4}{\theta\sqrt{\theta}} + \frac{2}{\sqrt{\theta}} \quad \text{①}$ $P'' = -4\left(\frac{-3}{2}\right)\theta^{-\frac{5}{2}} - \theta^{-\frac{3}{2}}$ $= 6\theta^{-\frac{5}{2}} - \theta^{-\frac{3}{2}}$ $= \frac{6}{\theta^2\sqrt{\theta}} - \frac{1}{\theta\sqrt{\theta}} \quad \text{①}$ | <ul style="list-style-type: none"> <li>Full marks were awarded if students correctly used the first derivative to determine maxima/ minima instead of the second derivative. However, if they did not show values for the first derivative around the stationary point marks were deducted.</li> <li>A few students used the quotient rule to differentiation instead of converting each term to index form before differentiating – this sometimes led to careless errors</li> <li>Marks were deducted if a check for maxima/ minima was not carried out.</li> <li>A few students unnecessarily calculated the perimeter once the value of <math>\theta</math> was determined</li> </ul> |

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|  |  | <p><b>(iii) cont.</b></p> <p><i>Stationary values : when <math>P' = 0</math></i></p> $P' = \frac{-4}{\theta\sqrt{\theta}} + \frac{2}{\sqrt{\theta}} = 0$ $\frac{-4 + 2\theta}{\theta\sqrt{\theta}} = 0$ $2\theta = 4$ $\theta = 2 \quad \text{①}$ <p><i>Nature of stationary value :</i></p> <p><i>When <math>\theta = 2</math></i></p> $P'' = \frac{6}{\theta^2\sqrt{\theta}} - \frac{1}{\theta\sqrt{\theta}}$ $= \frac{6}{4\sqrt{2}} - \frac{1}{2\sqrt{2}}$ $= \frac{4}{4\sqrt{2}}$ $> 0 \quad \text{①}$ <p><math>\therefore</math> Minimum perimeter when <math>\theta=2</math></p> |  |
|  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |

**Mathematics: Multiple Choice Answer Sheet**

|    |   |
|----|---|
|    |   |
| 1  | A |
| 2  | C |
| 3  | D |
| 4  | D |
| 5  | C |
| 6  | C |
| 7  | A |
| 8  | A |
| 9  | D |
| 10 | B |